

Assignment 8

Q1.Create an Employee and Manager program where Manager accesses Employee details using super keyword.

Code:

```
// Parent class
class Employee {
    String name = "Rahul";
    double salary = 50000;

    Employee() {
        System.out.println("Employee constructor called");
    }

    void displayDetails() {
        System.out.println("Employee Name: " + name);
        System.out.println("Employee Salary: " + salary);
    }
}

// Child class
class Manager extends Employee {
    String name = "Amit";
    double salary = 80000;

    Manager() {
        super(); // Calls parent constructor
        System.out.println("Manager constructor called");
    }

    void displayManagerDetails() {
        System.out.println("Parent Employee Name: " + super.name);
        System.out.println("Parent Employee Salary: " + super.salary);

        System.out.println("Manager Name: " + name);
        System.out.println("Manager Salary: " + salary);

        super.displayDetails(); // Calls parent method
    }
}
```

```
public class Main {  
    public static void main(String[] args) {  
        Manager manager = new Manager();  
        manager.displayManagerDetails();  
    }  
}
```

Output:

```
Employee constructor called  
Manager constructor called  
Parent Employee Name: Rahul  
Parent Employee Salary: 50000.0  
Manager Name: Amit  
Manager Salary: 80000.0  
Employee Name: Rahul  
Employee Salary: 50000.0
```

Q2. Develop a Vehicle Insurance System where child insurance classes access parent vehicle information using the super keyword.

Code:

```
// Parent class  
class Vehicle {  
    String vehicleNumber;  
    String vehicleType;
```

```

Vehicle(String vehicleNumber, String vehicleType) {
    this.vehicleNumber = vehicleNumber;
    this.vehicleType = vehicleType;
}

void displayVehicleDetails() {
    System.out.println("Vehicle Number: " + vehicleNumber);
    System.out.println("Vehicle Type: " + vehicleType);
}
}

// Child class
class CarInsurance extends Vehicle {
    String insuranceCompany;

    CarInsurance(String vehicleNumber, String vehicleType,
                  String insuranceCompany) {
        super(vehicleNumber, vehicleType); // Calls parent constructor
        this.insuranceCompany = insuranceCompany;
    }

    void displayInsuranceDetails() {
        System.out.println("Vehicle Insurance Details:");
        super.displayVehicleDetails(); // Calls parent method
        System.out.println("Insurance Company: " + insuranceCompany);
    }
}

public class Main {
    public static void main(String[] args) {
        CarInsurance car = new CarInsurance(
            "MH12AB1234",
            "Car",
            "ABC Insurance"
        );

        car.displayInsuranceDetails();
    }
}

```

Output:

Vehicle Insurance Details:

Vehicle Number: MH12AB1234

Vehicle Type: Car

Insurance Company: ABC Insurance